Lecture 12 (October 28th)

Remark. We continue with the final step of the Riemann mapping theorem. Let Σ be the set of all 1-1 analytic functions from D into U. We have proved if that $\phi \in \Sigma$ satisfies $\phi(D) \neq U$, then for every $z_0 \in D$ there is $\psi_0 \in \Sigma$ such that $|\psi_0'(z_0)| > |\phi'(z_0)|$.

(Step 3) If $z_0 \in D$ and $\eta_0 = \sup\{|\phi(z_0)| \mid \phi \in \Sigma\}$, then there is $h \in \Sigma$ such that $|h'(z_0)| = \eta_0$, which implies that $h(D) = U$.

Proof. We suggest that $\mathcal{A} = \{|\phi'(z_0) \mid \phi \in \Sigma\}$ is bounded. Suppose that it is not bounded. Then for every $n \in \boldsymbol{N}$, there is $\phi_n \in \Sigma$ such that $|\phi_n'(z_0)| > n$. However, notice that $|\phi_n(z)| < 1$ for all $z \in D$, $n \in \boldsymbol{N}$ and is uniformly bounded. By Montel's theorem, there is a subsequence $\{\phi_{n_k}\}$ which converges uniformly on every compact subset of D. Let $h(z) = \lim_{k\to\infty} \phi_{n_k}(z)$ for $z \in D$. Then, $h(z)$ is analytic in D and $\lim_{n\to\infty} \phi_{n_k}'(z) = h'(z)$ for every $z \in D$ so that $h'(z_0) = \lim_{k\to\infty} \phi_{n_k}'(z_0)$ which means that

$$|h'(z_0)| = \lim_{k\to\infty} |\phi_{n_k}'(z_0)| \geq \lim_{k\to\infty} n_k = \infty$$

which is a contradiction.

Knowing that $\eta_0 > 0$ (since $\phi' \neq 0$ for every $\phi \in \Sigma$) there is a subsequence $\{\phi_n\}$ in Σ such that $\lim_{n\to\infty} |\phi_n'(z_0)| = \eta_0$. Since $|\phi_n(z)| \leq 1$ for all $z \in D$, there is a subsequence $\{\phi_{n_k}\}$ which converges uniformly on all compact subsets of D to an analytic function h on D. Since $\lim_{k\to\infty} \phi_{n_k}'(z) = h'(z)$ for every $z \in D$, we have

$$|h(z_0)| = \lim_{k\to\infty} |\phi_{n_k}'(z_0)| = \eta_0$$

It remains to prove that $h \in \Sigma$. For every $z \in D$, $|h(z)| = \lim_{k\to\infty} |\phi_{n_k}(z)| \leq 1$ for all zin D as ϕ_{n_k} is a function from D to U. If $|h(z_1)| = 1$ for some $z_1 \in D$, by the maximum modulus theorem or the open mapping theorem, h is constant but $|h'(z_0)| = \eta_0 > 0$. This is again a contradiction. Thus, $h : D \to U$. Since ϕ_n is 1-1 analytic and $\phi_n \to h$ uniformly on all compact sets, we get that h is 1-1. $\square$

Example. Assume that $h(z_0) = 0$. If $h(z_0) = \beta \neq 0$ then $f = \phi_\beta \circ h \in \Sigma$ satisfies

$$|f'(z_0)| = |\phi_\beta'(\beta)h'(z_0)| = \frac{h'(z_0)}{1 - |\beta|^2} > |h'(z_0)|$$

Example. Prove or disprove whether for a $f : D \to U$ is analytic when $|f'(z_0)| \leq |h'(z_0)|$.

Proof. This is true. Define $g = f \circ h^{-1}$. Then, $g : U \to U$ is analytic.

$$g'(0) = f'(z_0)\big(h^{-1}\big)(0) = f'(z_0)\frac{1}{h'(z_0)} = \frac{f'(z_0)}{h'(z_0)}$$

since $|g'(0)| \leq 1$ which implies that $|f'(z_0)| \leq |h'(z_0)|$. $\square$

Theorem. If $g : U \to U$ is 1-1 onto analytic with $g(0) = 0$ and $g'(0) > 0$, then $g(z) = z$.

Theorem. If D is simply connected ($D \neq \boldsymbol{C}$) then for every $z_0 \in D$, there is a unique 1-1 analytic function h from D onto U satisfying $h(z_0) = 0$ and $h'(z_0) > 0$.

Proof. If $h \in \Sigma$ satisfies $|h'(z_0)| = \sup\{\phi'(z_0) \mid \phi \in \Sigma\}$, then $h(D) = U$ and $h(z_0) = 0$. If there exists f and g both satisfying the conditions above, define $h = f \circ g^{-1}$ and $h(0) = 0$ and $h'(0) = h'(z_0)/g'(z_0) > 0$ giving $h(z) = z$. $\square$

Lecture 13 (October 30th)

Theorem. Let $h'(z_0) = \sup\{|\phi'(z_0)| \mid \phi \in \Sigma\}$ then $h(\Omega) = U$ and $h \in \Sigma$. In addition, $h(z_0) = 0$. In addition, if we add the condition that $h'(z_0) > 0$, such h is uniquely determined.

Proof. If $h'(z_0) = re^{i\theta}$, then $f(z) = e^{-i\theta}h(z)$ satisfies $f \in \Sigma$, $f(\Omega) = U$, $f(z_0) = 0$ and $f'(z_0) = r > 0$. Let's prove uniqueness. If there exists f and g satisfying all conditions, then define

$$h(z) = f\Big(g^{-1}(z)\Big)$$

and $h : U \to U$ is 1-1 analytic and onto.

$$h'(0) = \frac{f'(z_0)}{g'(z_0)} > 0$$

and $h(0) = 0$. This implies that $h(z) = 0$ and that $f = g$. $\square$

Theorem. Let D be simply connected and $D \neq \boldsymbol{C}$. Then there is $\phi : D \leftarrow U$ that is 1-1 analytic and onto. Recall that if ϕ is 1-1, $\phi' \neq 0$ and is conformal. This tells us that there exists $\psi = \phi^{-1} : U \to D$ that is still 1-1 analytic and onto, thus conformal. Define $A(D)$ be the set of all analytic functions on D. Define $\Phi : A(D) \to A(U)$ by $\Phi(f) = f \circ \psi$. Then, Φ becomes a ring isomorphism.

Remark. Consider $\boldsymbol{C}^2 = \boldsymbol{C} \times \boldsymbol{C}$. The simplest domains in $\boldsymbol{C}^2$ are B_2 and U^2.

$$B_2 = \{z = (z_1, z_2) \in \boldsymbol{C}^2 \mid ||z|| < 1\} \qquad \& \qquad U^2 = U \times U$$

each called a bidisc and a unit ball. The norm $||z|| = \langle z, z\rangle^{1/2}$ is defined as

$$\langle z, z\rangle^{1/2} = \langle (z_1, z_2), (z_1, z_2)\rangle^{1/2} = \left(z_1\bar{z}_1 + z_2\bar{z}_2\right)^2 = \left(|z_1|^2 + |z_2|^2\right)^{1/2}$$

Theorem. (Residue theorem) If f is analytic inside and on a POSCC C except for isolated singularities at $z_1, \ldots, z_n$ inside C, then

$$\int_C f(z)\, dz = 2\pi i \sum_{k=1}^{n} \operatorname*{Res}_{z=z_k} f(z)$$

If $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ in $0 < |z - z_k| < \delta$ (Laurent series of f at z_k), then

$$a_{-1} = \operatorname*{Res}_{z=z_k} f(z)$$

If $\lim_{z\to z_0}(z - z_0)^k f(z_0) = A \neq 0$, we say that f has a pole of order k at z_0. If f has a simple pole (order 1) at z_0, then $\operatorname{Res}_{z=z_0} f(z) = \lim_{z\to z_0}(z - z_0)f(z)$. If

$$f(z) = \frac{p(z)}{q(z)}$$

(p, q analytic at z_0) has a simple pole at z_0 ($p(z_0) \neq 0$, $q(z_0) = 0$, and $q'(z_0) \neq 0$), then

$$\operatorname*{Res}_{z=z_0} f(z) = \lim_{z\to z_0}(z - z_0)\frac{p(z)}{q(z)} = \lim_{z\to z_0} \frac{p(z)}{\dfrac{q(z) - q(z_0)}{z - z_0}} = \frac{p(z_0)}{q'(z_0)}$$

Example. Evaluate for $0 < \alpha < 1$,

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1 + e^x}\, dx$$

Proof. We need to take a rectangular contour C_R from $-R$ to R and height $2\pi i$ notice that the above function has a residue at πi. Take

$$f(z) = \frac{\alpha z}{1 + e^z}$$

For every $R > 0$,

$$\int_{C_R} f(z)\, dz = 2\pi i \operatorname*{Res}_{z=\pi i} f(z) = 2\pi i \frac{e^{\alpha\pi i}}{e^{\pi i}} = -2\pi i e^{\alpha\pi i}$$

On the other hand,

$$\int_{C_R} f(z)\,dz = \int_{-R}^{R} \frac{e^{\alpha x}}{1+e^x}\,dx + \int_0^{2\pi} \frac{e^{\alpha(R+iy)}}{1+e^{R+iy}} i\,dy + \int_R^{-R} \frac{e^{\alpha(x+2\pi i)}}{1+e^{x+2\pi i}}\,dx + \int_{2\pi}^{0} \frac{e^{\alpha(-R+iy)}}{1+e^{-R+iy}} i\,dy$$

The second term goes to zero as $R \to \infty$ and

$$-2\pi i e^{\alpha\pi i} = \int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x}\,dx \left[1 - e^{2\alpha\pi i}\right]$$

resulting in

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x}\,dx = -\frac{2\pi i}{1-e^{2\alpha\pi i}} = \frac{\pi}{\dfrac{e^{\alpha\pi i} - e^{-\alpha\pi i}}{2i}} = \frac{\pi}{\sin\alpha\pi}$$

If we further put $t = e^x$, $dx = e^{-x}$, and $dt/t = dx$. We therefore have

$$\int_0^{\infty} \frac{t^{\alpha-1}}{1+t}\,dt = \frac{\pi}{\sin\alpha\pi}$$

□

Lecture 14 (November 4th)

Definition. (Fourier transform of f on $\boldsymbol{R}$) The Fourier transform of f on $\boldsymbol{R}$ is defined as

$$\widehat{f}(t) = \mathcal{F}(f)(t) = \int_{-\infty}^{\infty} f(x) e^{-itx}\,dx$$

then

$$f(x) \sim \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f}(t) e^{itx}\,dt$$

for a very huge class of functions f, the above is an equality. If we define

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x t}\,dx$$

then

$$f(x) \sim \int_{-\infty}^{\infty} \hat{f}(t) e^{2\pi i x t}\,dt$$

(this still works for f on $\boldsymbol{R}^n$) We also conventionally use

$$\widehat{f}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-itx}\,dx$$

then

$$f(x) \sim \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{f}(t) e^{ixt}\,dt$$

Lastly, we also use

$$\widehat{f}(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(x)e^{-itx}\,dx$$

then

$$f(x) \sim \int_{-\infty}^{\infty} \hat{f}(t)e^{ixt}\,dx$$

The above works for $f \in L^1$, meaning that $\int_{-\infty}^{\infty} |f(x)|\,dx < \infty$. We can also define the above as

$$\widehat{f}(t) = \lim_{A\to\infty}\int_{-A}^{A} f(x)e^{-itx}\,dx$$

when the limits exists.

Example. (Example of transform not being L^1) Consider the L^1 function

$$f(x) = \begin{cases} 1 & -1 < x < 1 \\ 0 & |x| > 1 \end{cases}$$

the Fourier transform is then

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x)e^{-itx}\,dx = \int_{-1}^{1} e^{-itx}\,dx = -\frac{1}{it}e^{-itx}|_0^1 = -\frac{1}{it}\left[e^{-it} - e^{it}\right] = \frac{2\sin t}{t}$$

(which is 2 at $t = 0$) Notice that

$$\int_0^{N\pi} \left|\frac{\sin t}{t}\right| dt = \sum_{k=0}^{N-1} \int_{k\pi}^{(k+1)\pi} \left|\frac{\sin t}{t}\right| dt$$

where

$$\int_{k\pi}^{(k+1)\pi} \left|\frac{\sin t}{t}\right| dt \geq \int_{k\pi+\frac{\pi}{6}}^{k\pi+\frac{5\pi}{6}} \left|\frac{\sin t}{t}\right| dt \geq \frac{1}{2(k+1)\pi}\frac{2\pi}{3} = \frac{1}{3(k+1)}$$

and that the result is not L^1. Is the inverse transform of $g(t) = 2\sin/t$

$$\lim_{A\to\infty}\int_{-A}^{A} g(t)e^{ixt}\,dt = f(x)$$

is a good question.

Example. Let $a > 0$ and consider the L^1 function

$$f(x) = \frac{1}{x^2 + a^2}$$

and

$$\hat{f}(t) = \int_{-\infty}^{\infty} \frac{e^{-itx}}{x^2 + a^2}\,dx$$

Proof. Define

$$f(z) = \frac{e^{-itz}}{z^2 + a^2}$$

and use the residue theorem. In the upper hemisphere contour C_R with $R > a$,

$$\int_{C_R} f(z)\,dz = 2\pi i \operatorname*{Res}_{z=ai} f(z) = 2\pi i \frac{e^{-it(ai)}}{2ai} = \frac{\pi}{a} e^{at}$$

Recall that for a simple pole,

$$\operatorname*{Res}_{z=z_0} \left(\frac{p(z)}{q(z)}\right) = \frac{p(z_0)}{q'(z_0)}$$

The above is equal to

$$\int_{-R}^{R} \frac{e^{-itx}}{x^2 + a^2}\,dx + \int_0^{\pi} \frac{e^{-itRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} iRe^{i\theta}\,d\theta$$

The absolute value of the numerator is

$$\left|e^{-itRe^{i\theta}}\right| = \left|e^{-it(R\cos\theta + iR\sin\theta)}\right| = e^{tR\sin\theta}$$

and the second term of the above vanishes at $R \to \infty$ if the above is bounded and $t \leq 0$. In the case that $t > 0$, we take the lower hemisphere and the pole is at $-ai$.

$$\int_{C_R} f(z)\,dz = -2\pi i \operatorname*{Res}_{z=ai} f(z) = 2\pi i \frac{e^{-it(-ai)}}{-2ai} = \frac{\pi}{a} e^{-at}$$

Likewise,

$$\int_{C_R} f(z)\,dz = \int_{-R}^{R} \frac{e^{-itx}}{x^2 + a^2}\,dx + \int_0^{-\pi} \frac{e^{-itRe^{i\theta}}}{R^2 e^{2i\theta} + a^2} iRe^{i\theta}\,d\theta$$

and

$$\left|e^{-itRe^{i\theta}}\right| = e^{tR\sin\theta} < 1$$

since $-\pi\theta < \theta < 0$. Thus,

$$\hat{f}(t) = \int_{-\infty}^{\infty} \frac{e^{-itx}}{x^2 + a^2}\,dx = \frac{\pi}{a} e^{-a|t|}$$

which is L^1. Let $a = 1$ and try to invert the above. If $g(t) = \pi e^{-|t|}$ then

$$
\begin{aligned}
\frac{1}{2\pi}\int_{-\infty}^{\infty} g(t)e^{ixt}\,dt =& \frac{1}{2}\int_{-\infty}^{\infty} e^{-|t|}e^{ixt}\,dt = \frac{1}{2}\Big[\int_{-\infty}^{0} e^{t}e^{ixt}\,dt + \int_{0}^{\infty} e^{-t}e^{ixt}\,dt\Big] \\
=& \frac{1}{2}\Big[\int_{-\infty}^{0} e^{(1+ix)t}\,dt + \int_{0}^{\infty} e^{(-1+ix)t}\,dt\Big] \\
=& \frac{1}{2}\Big[\frac{e^{(1+ix)t}}{1+ix}\Big]_{t=-\infty}^{0} + \frac{1}{2}\Big[\frac{e^{(-1+ix)t}}{-1+ix}\Big]_{t=0}^{\infty} \\
=& \frac{1}{2}\Big[\frac{1}{1+ix}\Big] + \frac{1}{2}\Big[\frac{1}{1-ix}\Big] \\
=& \frac{1}{2}\Big(\frac{1}{1+ix} + \frac{1}{1-ix}\Big) = \frac{1}{1+x^2}
\end{aligned}
$$

□

Example. Let $f(x) = e^{-x^2/2}$. Then,

$$
\begin{aligned}
\hat{f}(t) &= \int_{-\infty}^{\infty} f(x)e^{-itx}\,dx = \int_{-\infty}^{\infty} e^{-x^2/2}e^{-itx}\,dx \\
&= \int_{-\infty}^{\infty} e^{-(x^2+2itx)}\,dx = \Big[\int_{-\infty}^{\infty} e^{-(x+it)^2/2}\,dx\Big]e^{-t^2/2}
\end{aligned}
$$

which is $\sqrt{2\pi}$ when $t = 0$. When $t > 0$, we take the rectangular contour with base $2R$ and height $x + it$. By Cauchy,

$$
0 = \int_{C_R} e^{-z^2/2}\,dz = \int_{-R}^{R} e^{-x^2/2}\,dx + (\mathrm{II}) + (\mathrm{III}) + (\mathrm{IV})
$$

We see that

$$
(\mathrm{III}) = \int_{R}^{-R} e^{-(x+it)^2/2}\,dt
$$

and

$$
(\mathrm{II}) = \int_{0}^{t} e^{(R+iy)^2/2} i\,dy
$$

We now use the fact that the absolute value of (II) and (III) goes to zero as $R \to \infty$. We finally have

$$
\int_{-\infty}^{\infty} e^{-(x+it)^2}\,dx = \sqrt{2\pi}
$$

with

$$
\hat{f}(t) = \sqrt{2\pi}e^{-t^2/2}
$$

Lecture 15 (November 6th)

Recall. We have previously shown the result (for $0 < \alpha < 1$)

$$\int_{-\infty}^{\infty} \frac{e^{\alpha x}}{1+e^x}\, dx = \frac{\pi}{\sin \alpha\pi}$$

where the change of variables $x = t^\alpha$ gives

$$\int_0^{\infty} \frac{x^{\alpha-1}}{1+x}\, dx = \frac{\pi}{\sin \alpha\pi}$$

Now put $x = t^\beta$ for $0 < \beta < \infty$.

$$\int_0^{\infty} \frac{x^{\alpha-1}}{1+x}\, dx = \int_0^{\infty} \frac{t^{\alpha\beta-\beta}}{1+t^\beta} \beta t^{\beta-1}\, dt = \beta \int_0^{\infty} \frac{t^{\alpha\beta-1}}{1+t^\beta}\, dt = \frac{\pi}{\alpha\pi}$$

Take $\alpha = 1/\beta$ such that $\beta > 1$. We get

$$\int_0^{\infty} \frac{1}{1+t^\beta}\, dt = \frac{1}{\beta \sin \dfrac{\pi}{\beta}}$$

for $\beta > 1$.

Recall. We have considered Fourier transform of

$$f(x) = \begin{cases} 1 & -1 < x < 1 \\ 0 & |x| > 1 \end{cases}$$

which was 2 times the snc function

$$\hat{f}(t) = \int_{-\infty}^{\infty} f(x) e^{-itx}\, dx = \frac{2\sin t}{t}$$

We want to check that the inverse transform properly gives the original function.

Example. Find $\lim_{R\to\infty} \int_{-R}^{R} \dfrac{\sin x}{x} e^{itx}\, dx$. Notice that the kernel is a entire function, and that

$$\int_{-R}^{R} f(z)\, dz = \int_{C_R} f(z)\, dz = \int_{C_R} \frac{e^{iz} - e^{-iz}}{2iz} e^{itz}\, dz$$

where C_R is defined as contour the avoids the origin. Then,

$$\begin{aligned} \int_{-R}^{R} f(z)\, dz = & \int_{C_R} \frac{1}{2i} \Big[\frac{e^{i(t+1)z} - e^{i(t-1)z}}{z} \Big]\, dz \\ = & \frac{1}{2i} \Big[\int_{C_R} \frac{e^{i(t+1)z}}{z}\, dz - \int_{C_R} \frac{e^{i(t-1)z}}{z}\, dz \Big] \\ = & \varphi_R(t+1) - \varphi_R(t-1) \end{aligned}$$

where $\varphi_R(t) = \dfrac{1}{2i}\int_{C_R} \dfrac{e^{itz}}{z}\,dz$. Notice that

$$\left|\frac{e^{itz}}{z}\right| = \left|\frac{e^{itRe^{i\theta}}}{Re^{i\theta}} iRe^{i\theta}\right| = |e^{-tR\sin\theta}|$$

To evaluate

$$\varphi_R(t) = \frac{1}{2i}\int_{C_R} \frac{e^{itz}}{z}\,dz$$

where for $t < 0$, we choose a POSCC so that θ is from 0 to $-\pi$. We find

$$\lim_{R\to\infty}\int_{C_R} \frac{1}{2i}\frac{e^{itz}}{z}\,dz = 0$$

For $t > 0$, we choose the POSCC $\Gamma_R = C_R \cup \{\text{upper semi circle}\}$. We find

$$\int_{\Gamma_R} \frac{e^{itz}}{z}\,dz = 2\pi i$$

thus

$$\lim_{R\to\infty}\int_{C_R} \frac{1}{2i}\frac{e^{itz}}{z}\,dz = \pi$$

Our goal was

$$\begin{aligned}
\lim_{R\to\infty}\int_{-R}^{R} \frac{\sin x}{x}e^{itx}\,dx = \lim_{R\to\infty}\int_{C_R} \frac{\sin x}{x}e^{itx}\,dx &= \lim_{R\to\infty}\Big[\varphi_R(t+1) - \varphi_R(t-1)\Big] \\
&= \lim_{R\to\infty}\varphi_R(t+1) - \lim_{R\to\infty}\varphi_R(t-1) \\
&= \begin{cases} \pi & -1 < t < 1 \\ \pi/2 & t = 0 \\ 0 & |t| > 1 \end{cases}
\end{aligned}$$

Now, $\hat{f}(t) = 2\sin t/t$ gives

$$\lim_{R\to\infty}\int_{-R}^{R} \hat{f}(t)e^{ixt}\,dt$$

We can further find that $\lim_{R\to\infty}\varphi(0) = (i\pi)/2i = \pi/2$.

Theorem. (Big theorem)

(i) If f and f' are L^1, $\hat{f}'(t) = it\hat{f}(t)$

(ii) If $f(x)$ and $xf(x)$ are in L^1, then

$$\big(xf(x)\big)\hat{}\,(t) = i\big(\hat{f}\big)'(t)$$

and

$$\mathcal{F}\big(xf(x)\big)(t) = i\big(\hat{f}\big)'(t)$$

Proof. (I)

$$\hat{f'}(t) = \int_{-\infty}^{\infty} f'(x)e^{-itx}\,dx = \Big[f(x)e^{-itx}\Big]_{x=-\infty}^{\infty} + \int_{-\infty}^{\infty} f(x)ite^{-itx}\,dx$$

However, the first term is zero as

$$\lim_{x\to\infty} f(x) = f(0) + \int_0^{\infty} f'(x)\,dx$$

where $\int_0^\infty f'(x)\,dx = \lim_{x\to\infty} f(x) - f(0)$ and if $\lim_{x\to\infty} f(x) = \alpha$, $\alpha = 0$. (II)

$$\frac{d}{dt}\hat{f}(x) = \frac{d}{dt}\int_{-\infty}^{\infty} f(x)e^{-ixt}\,dt = \int_{-\infty}^{\infty} -ixf(x)e^{-ixt}\,dt = -i\widehat{\Big[xf(x)\Big]}(t)$$

□

Example. Last class we saw that for $f(x) = \exp(-x^2/2)$ implies $\hat{f}(t) = \sqrt{2\pi}\exp(-t^2/2)$. Then,

$$f'(x) = -xe^{-x^2/2}$$

so that

$$f'(x) = -xf(x)$$

Applying the Fourier transform,

$$it\hat{f}(t) = -i\big(\hat{f}\big)'(t)$$

and

$$\big(\hat{f}\big)'(t) = -t\hat{f}(t)$$